UNIVERSIDAD DE HUELVA

MÁSTER EN INGENIERÍA INFORMÁTICA

Universidad de Huelva
Escuela Técnica Superior de Ingeniería

Tarea: PaaS

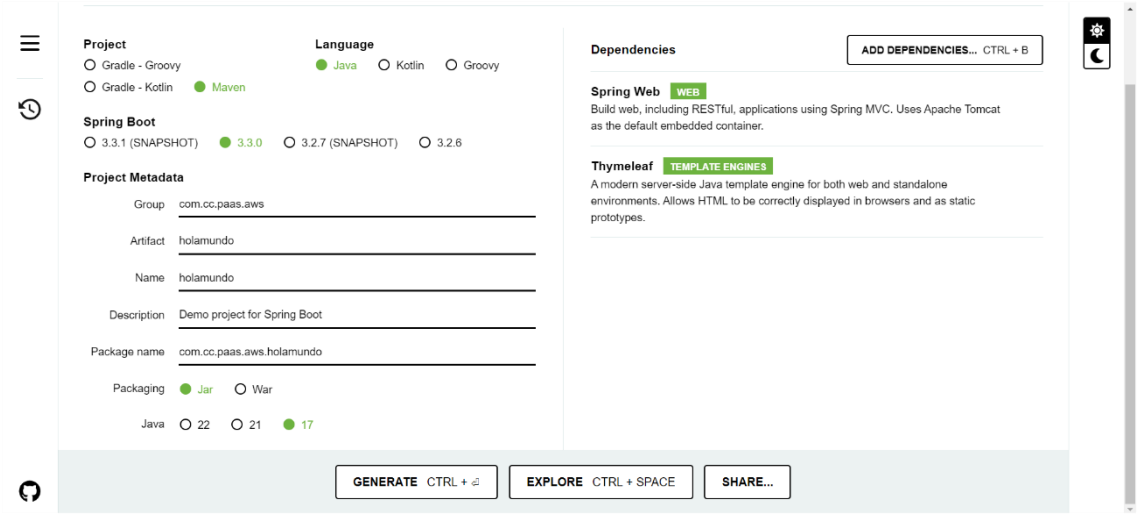
Ciclo lectivo 2023-24

Alumno: Jamal Habboula

Fecha de Entrega: 02/06/2024

Crear una aplicación Spring Boot usando Spring Initializr – AWS

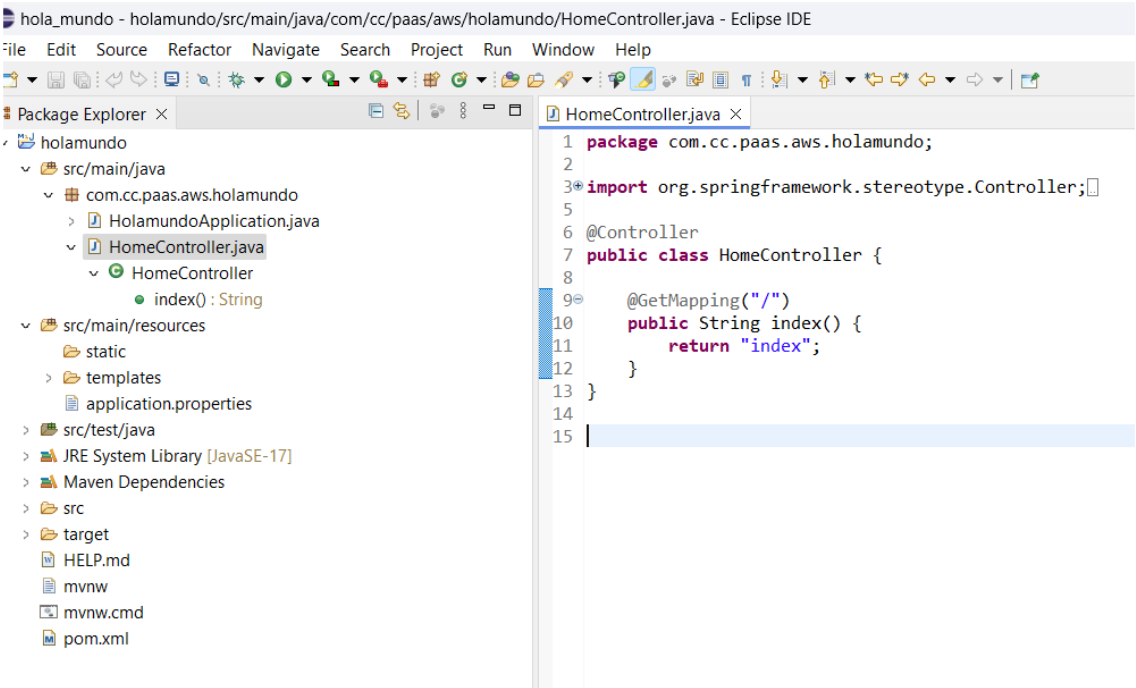
Configurar Spring Initializr



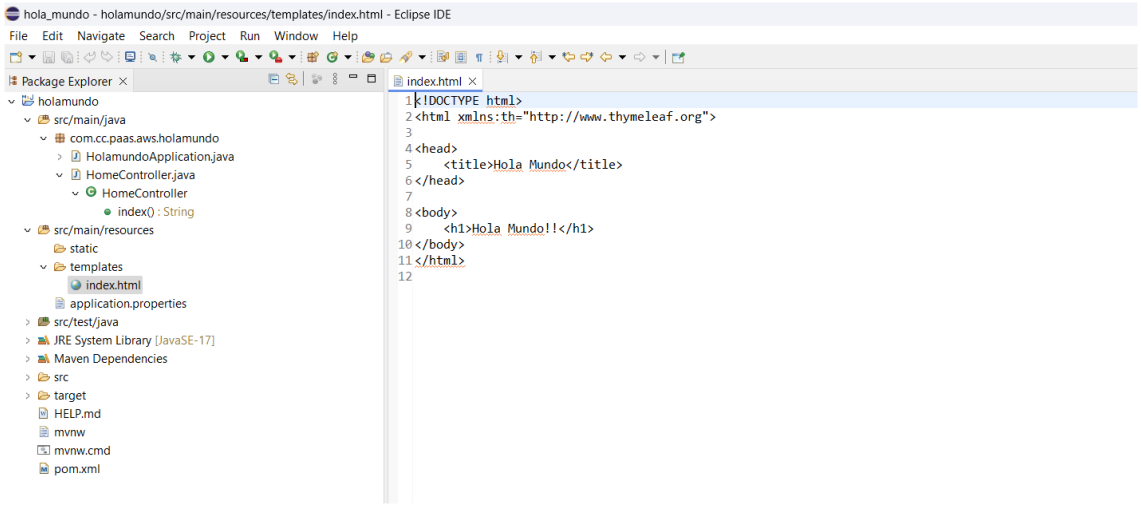
Genera el proyecto, guárdalo en local y súbelo y descomprime en mi IDE (Eclipse IDE for Java Developers - 2024-03)

- Crear un Controlador y una Vista

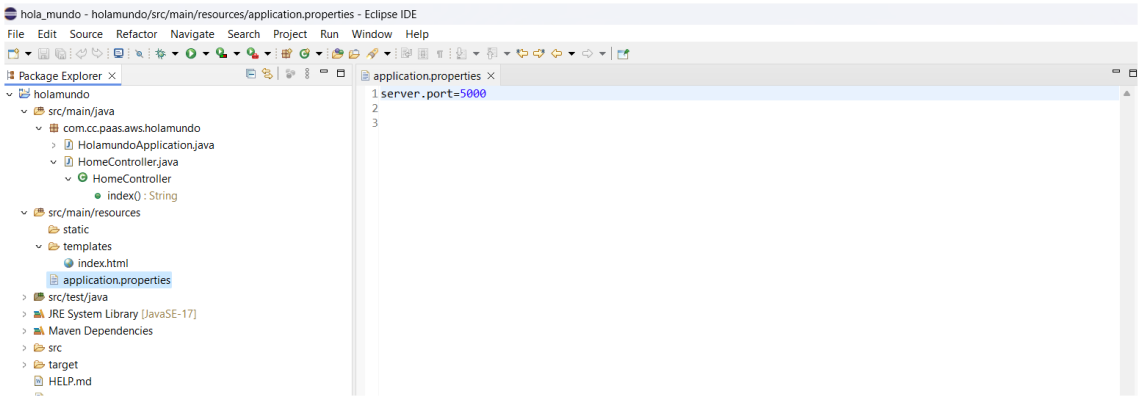
Controlador HomeController.java



Vista index.html



- Cambiar el Puerto de la Aplicación al 5000

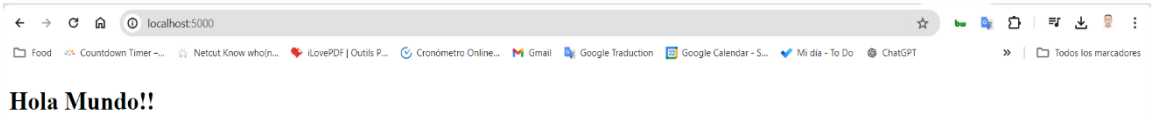


- Navegar al directorio de mi proyecto utilizando comando ‘mvn spring-boot:run’ (Símbolo del sistema de Windows)

```
Símbolo del sistema - mvn sp
Microsoft Windows [Versión 10.0.22631.3672]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\jamal>cd C:\Users\jamal\Downloads\hola_mundo\holamundo

C:\Users\jamal\Downloads\hola_mundo\holamundo>mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:holamundo >------------------------
[INFO] Building holamundo 0.0.1-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] >>> spring-boot:2.7.3:run (default-cli) > test-compile @ holamundo >>>
[INFO]
[INFO] --- resources:3.2.0:resources (default-resources) @ holamundo ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Using 'UTF-8' encoding to copy filtered properties files.
[INFO] Copying 1 resource
[INFO] Copying 1 resource
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ holamundo ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.2.0:testResources (default-testResources) @ holamundo ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Using 'UTF-8' encoding to copy filtered properties files.
[INFO] skip non existing resourceDirectory C:\Users\jamal\Downloads\hola_mundo\holamundo\src\test\resources
```

```
[INFO] Attaching agents: []

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v2.7.3)

2024-06-02 15:00:59.259  INFO 22004 --- [           main] c.c.p.a.holamundo.HolamundoApplication   : Starting HolamundoA
pplication using Java 17.0.11 on JAMALPC with PID 22004 (C:\Users\jamal\Downloads\hola_mundo\holamundo\target\classes st
arted by jamal in C:\Users\jamal\Downloads\hola_mundo\holamundo)
2024-06-02 15:00:59.262  INFO 22004 --- [           main] c.c.p.a.holamundo.HolamundoApplication   : No active profile s
et, falling back to 1 default profile: "default"
2024-06-02 15:00:59.872  INFO 22004 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized
with port(s): 5000 (http)
2024-06-02 15:00:59.883  INFO 22004 --- [           main] o.apache.catalina.core.StandardService   : Starting service [T
omcat]
2024-06-02 15:00:59.883  INFO 22004 --- [           main] org.apache.catalina.core.StandardEngine  : Starting Servlet en
gine: [Apache Tomcat/9.0.65]
2024-06-02 15:01:00.000  INFO 22004 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring
 embedded WebApplicationContext
2024-06-02 15:01:00.001  INFO 22004 --- [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplication
Context: initialization completed in 698 ms
2024-06-02 15:01:00.307  INFO 22004 --- [           main] o.s.b.a.w.s.WelcomePageHandlerMapping    : Adding welcome page
 template: index
2024-06-02 15:01:00.484  INFO 22004 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on p
ort(s): 5000 (http) with context path ''
2024-06-02 15:01:00.491  INFO 22004 --- [           main] c.c.p.a.holamundo.HolamundoApplication   : Started HolamundoAp
```

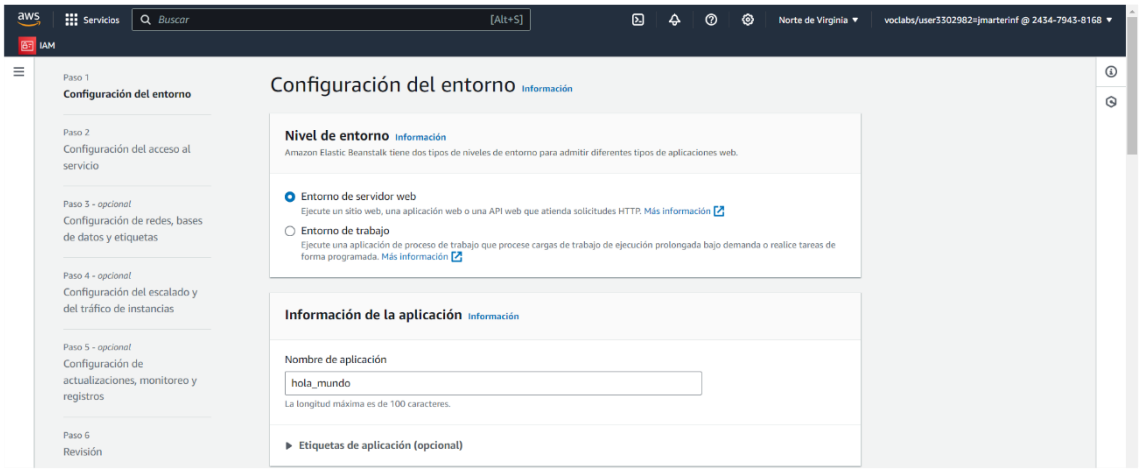
- Ingresar la URL para acceder a mi aplicación http://localhost:5000/

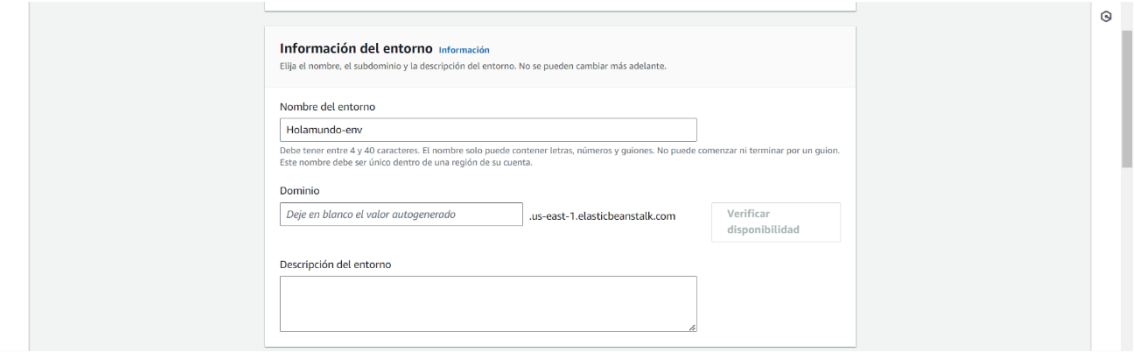


Desplegar una aplicación en AWS Elastic Beanstalk utilizando la consola web de AWS:

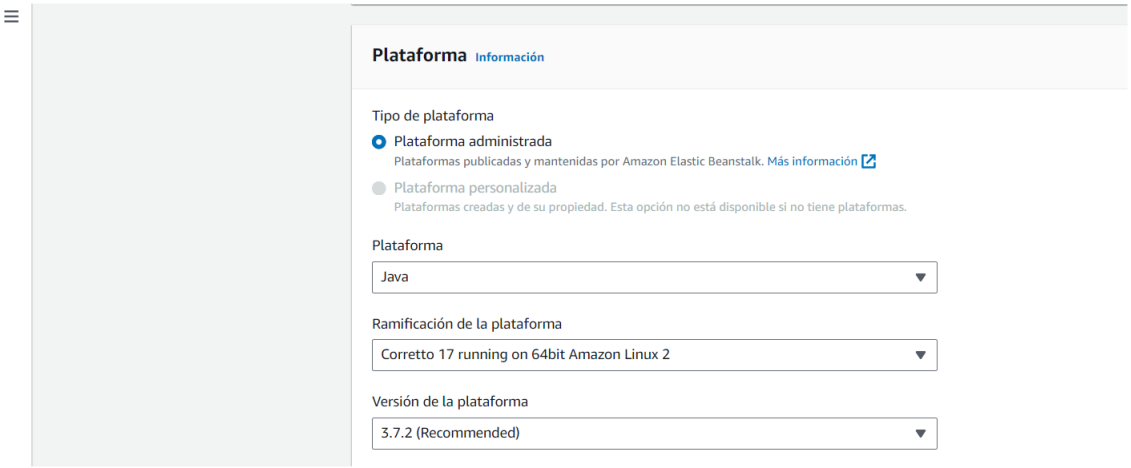
- Crear Aplicación:

Configuración del Entorno

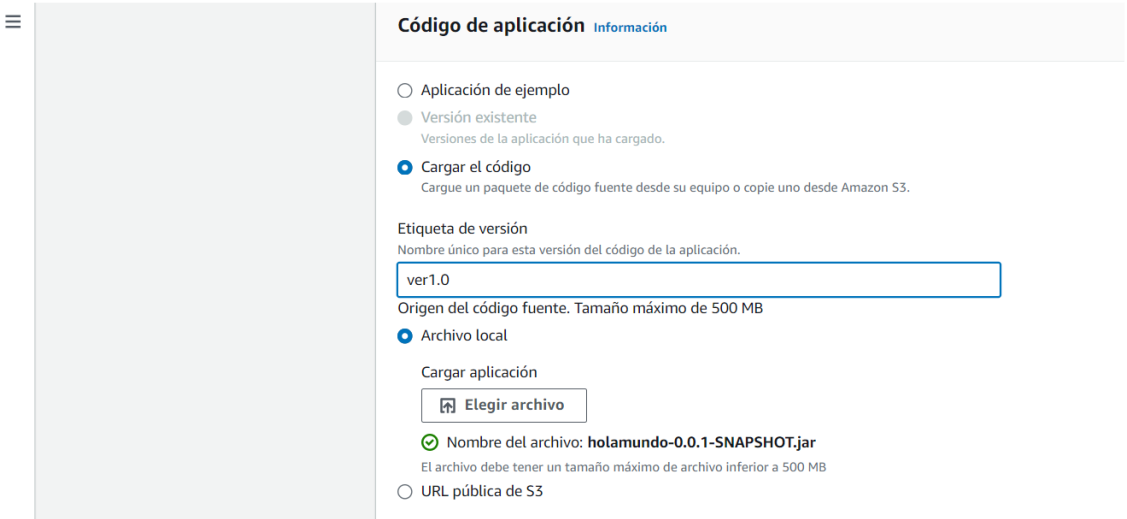




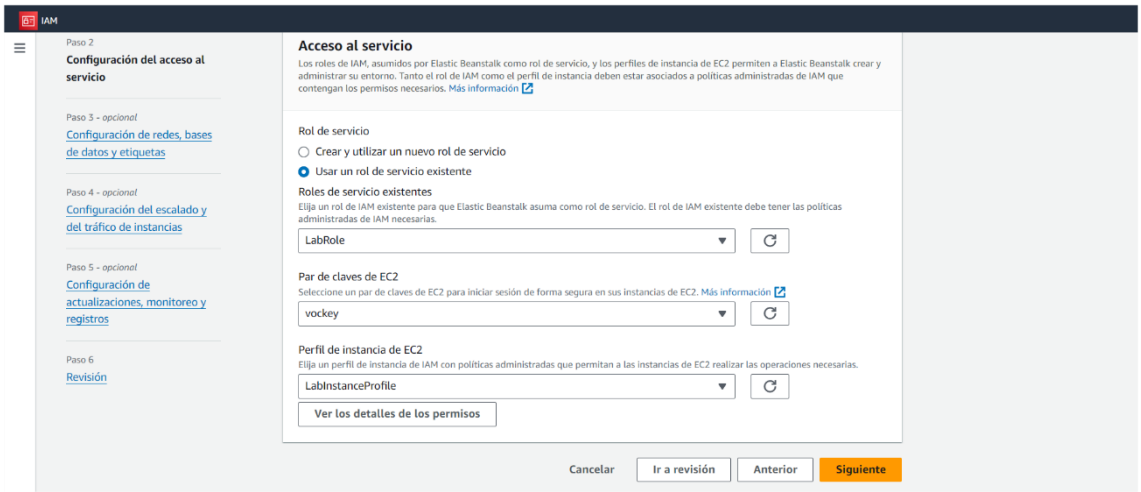
Plataforma



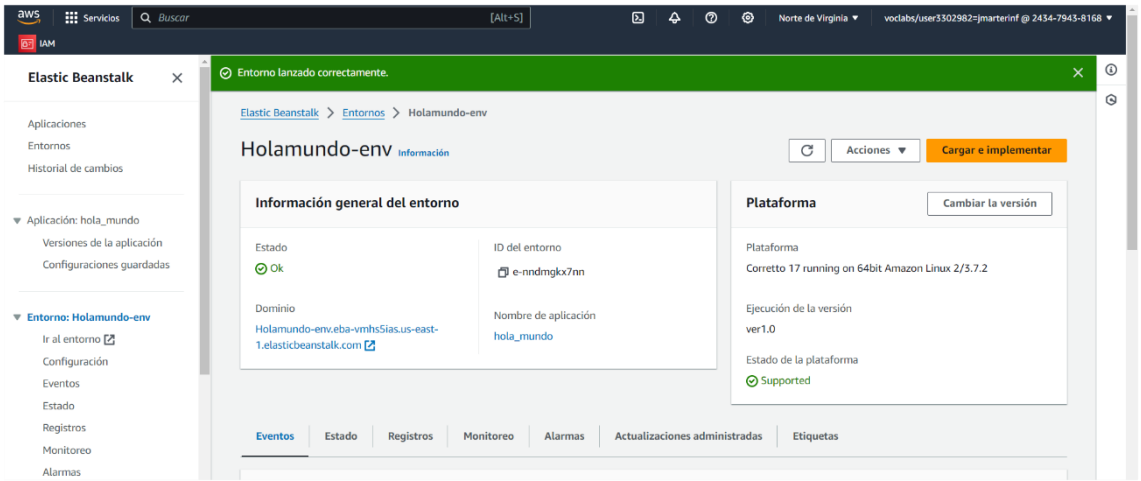
Código de Aplicación



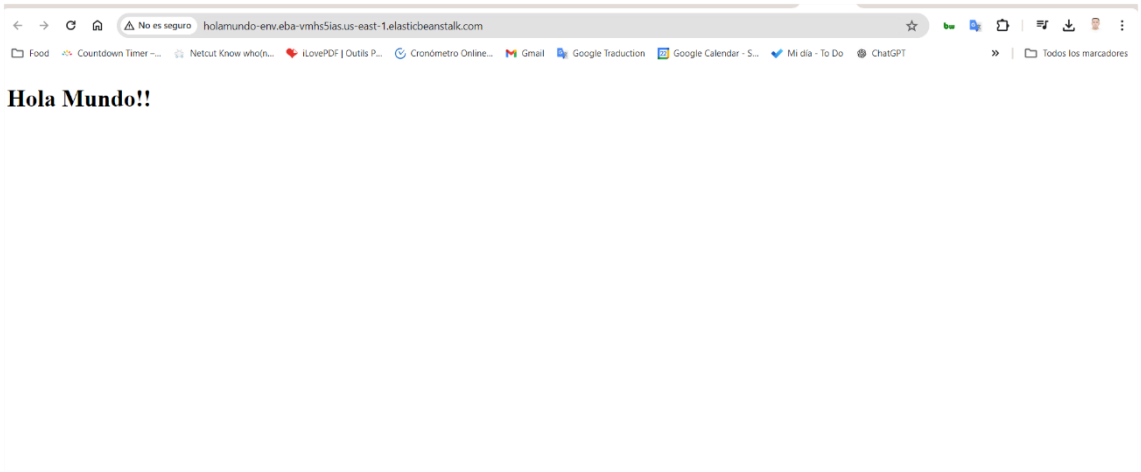
Configuración de Acceso al Servicio



Comprobar la Aplicación

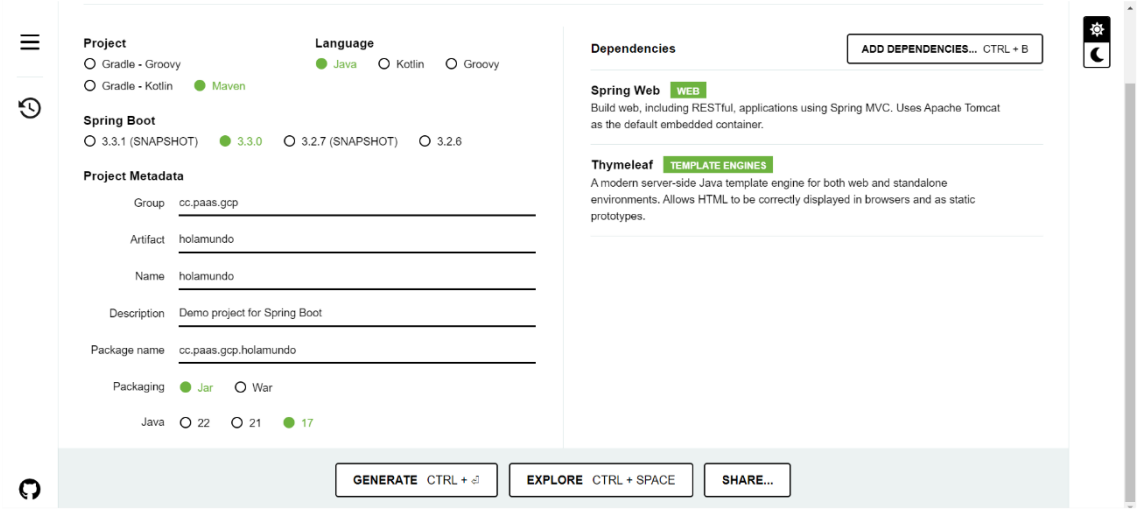


Acceder a la URL generada por Elastic Beanstalk 'http://holamundo-env.eba-vmhs5ias.us-east-1.elasticbeanstalk.com/'



Crear una aplicación Spring Boot usando Spring Initializr – GCP

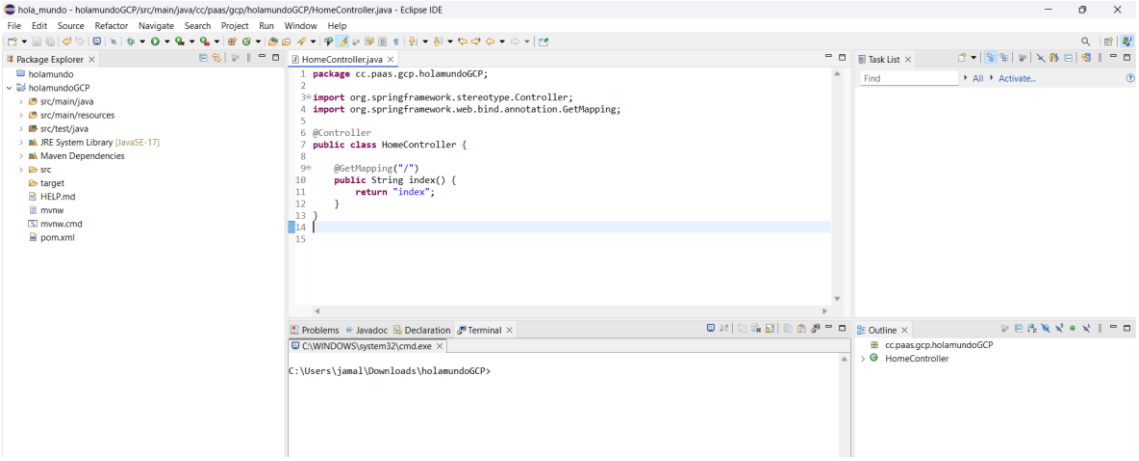
Configurar Spring Initializr



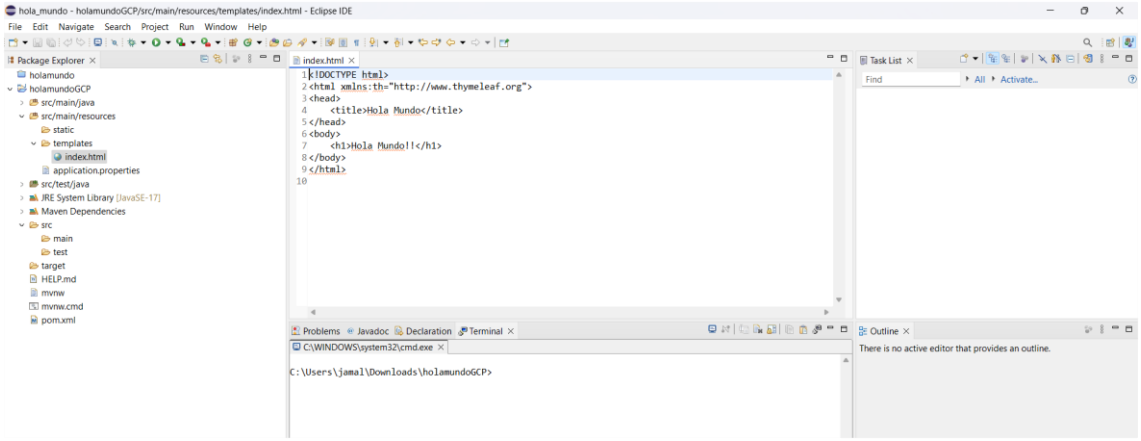
Genera el proyecto, guárdalo en local y súbelo y descomprime en mi IDE (Eclipse IDE for Java Developers - 2024-03)

- Crear un Controlador y una Vista

Controlador HomeController.java



Vista index.html



Navegar al directorio de mi proyecto utilizando comando ‘mvn spring-boot:run’ (Símbolo del sistema de Windows)

```
Microsoft Windows [Versión 10.0.22631.3672]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\jamal>cd C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP

C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP>mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< cc.paas.gcp:holamundoGCP >-----------------------
[INFO] Building holamundoGCP 0.0.1-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] >>> spring-boot:3.3.0:run (default-cli) > test-compile @ holamundoGCP >>>
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ holamundoGCP ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ holamundoGCP ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ holamundoGCP ---
[INFO] skip non existing resourceDirectory C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ holamundoGCP ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:3.3.0:run (default-cli) < test-compile @ holamundoGCP <<<
[INFO]
```

```
Símbolo del sistema - mvn sp

  .   ____          _            __ _ _
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( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v3.3.0)

2024-06-02T17:36:18.298+02:00  INFO 15464 --- [holamundoGCP] [           main] c.p.g.h.HolamundoGcpApplication
: Starting HolamundoGcpApplication using Java 17.0.11 with PID 15464 (C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP
\target\classes started by jamal in C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP)
2024-06-02T17:36:18.301+02:00  INFO 15464 --- [holamundoGCP] [           main] c.p.g.h.HolamundoGcpApplication
: No active profile set, falling back to 1 default profile: "default"
2024-06-02T17:36:18.950+02:00  INFO 15464 --- [holamundoGCP] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer
: Tomcat initialized with port 8080 (http)
2024-06-02T17:36:18.965+02:00  INFO 15464 --- [holamundoGCP] [           main] o.apache.catalina.core.StandardService
: Starting service [Tomcat]
2024-06-02T17:36:18.965+02:00  INFO 15464 --- [holamundoGCP] [           main] o.apache.catalina.core.StandardEngine
: Starting Servlet engine: [Apache Tomcat/10.1.24]
2024-06-02T17:36:19.015+02:00  INFO 15464 --- [holamundoGCP] [           main] o.a.c.c.C.[Tomcat].[localhost].[/]
: Initializing Spring embedded WebApplicationContext
2024-06-02T17:36:19.016+02:00  INFO 15464 --- [holamundoGCP] [           main] w.s.c.ServletWebServerApplicationContext
: Root WebApplicationContext: initialization completed in 674 ms
2024-06-02T17:36:19.128+02:00  INFO 15464 --- [holamundoGCP] [           main] o.s.b.a.w.s.WelcomePageHandlerMapping
: Adding welcome page template: index
2024-06-02T17:36:19.330+02:00  INFO 15464 --- [holamundoGCP] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer
: Tomcat started on port 8080 (http) with context path '/'
2024-06-02T17:36:19.336+02:00  INFO 15464 --- [holamundoGCP] [           main] c.p.g.h.HolamundoGcpApplication
: Started HolamundoGcpApplication in 1.691 seconds (process running for 1.947)
```
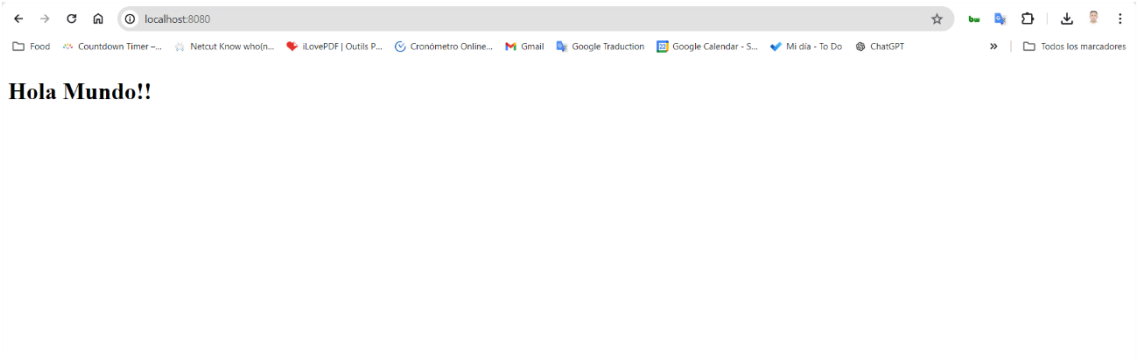
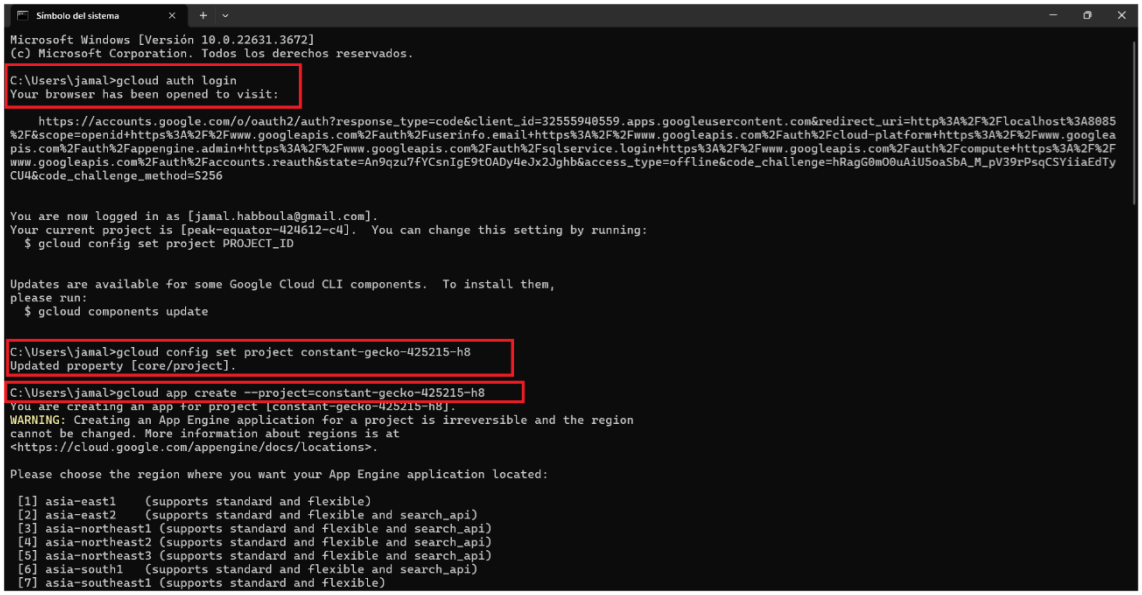

- Ingresar la URL para acceder a mi aplicación http://localhost:8080/



Desplegar en Google App Engine

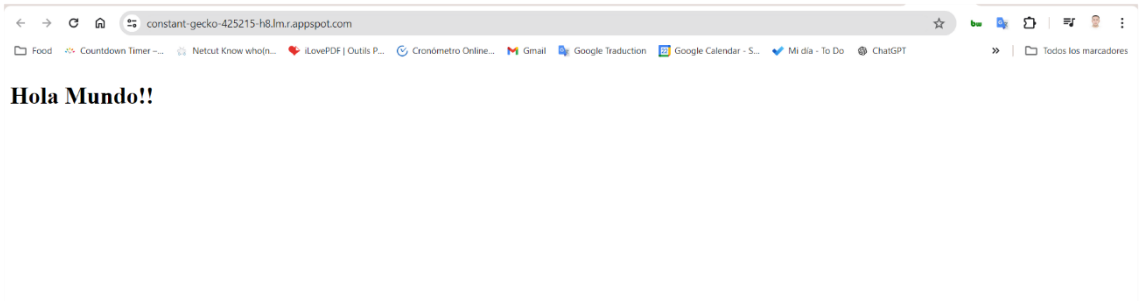
Iniciar sesión en Google Cloud y seleccionar el Proyecto y crea el entorno en App Engine



Desplegar la Aplicación:

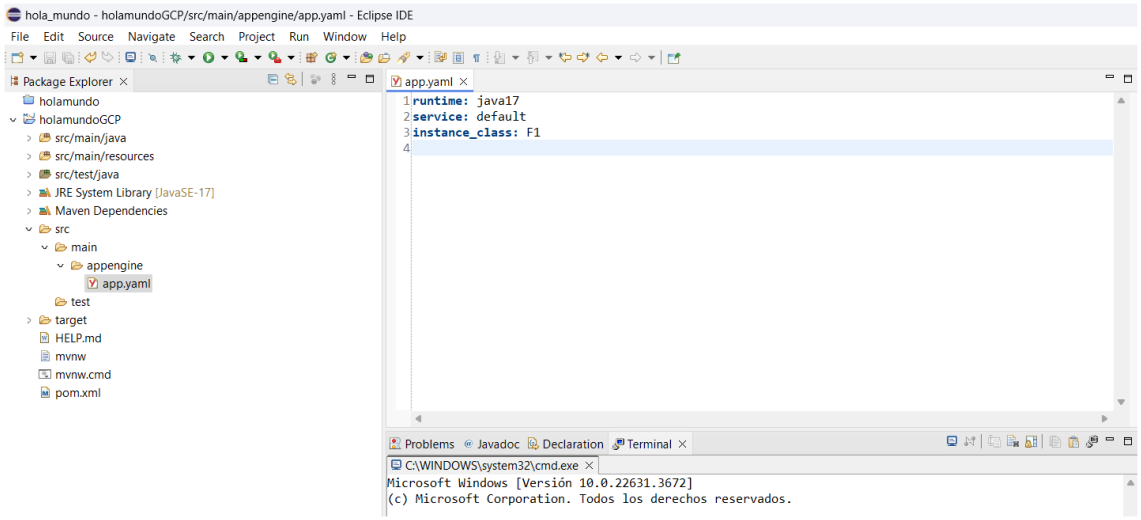
```
 [22] us-west3      (supports standard and flexible and search_api)
 [23] us-west4      (supports standard and flexible and search_api)
 [24] cancel
Please enter your numeric choice:  10

Creating App Engine application in project [constant-gecko-425215-h8] and region [europe-central2]....done.
Success! The app is now created. Please use `gcloud app deploy` to deploy your first app.

C:\Users\jamal>cd C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP

C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP>gcloud app deploy
Services to deploy:

descriptor:                  [C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP\pom.xml]
source:                      [C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP]
target project:              [constant-gecko-425215-h8]
target service:              [default]
target version:              [20240602t175838]
target url:                  [https://constant-gecko-425215-h8.lm.r.appspot.com]
target service account:      [constant-gecko-425215-h8@appspot.gserviceaccount.com]


Do you want to continue (Y/n)?  y

Beginning deployment of service [default]...
Created .gcloudignore file. See `gcloud topic gcloudignore` for details.
#=================================================================#
#= Uploading 16 files to Google Cloud Storage                    =#
#=================================================================#
File upload done.
Updating service [default]...done.
Setting traffic split for service [default]...done.
Deployed service [default] to [https://constant-gecko-425215-h8.lm.r.appspot.com]

You can stream logs from the command line by running:
  $ gcloud app logs tail -s default

To view your application in the web browser run:
  $ gcloud app browse
```

Ingresar la URL para acceder a mi aplicación https://constant-gecko-425215-h8.lm.r.appspot.com/



Crear un Nuevo Servicio

Crear el Archivo de Configuración app.yaml



Desplegar el Nuevo Servicio

```
Símbolo del sistema

C:\Users\jamal>cd C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP

C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP>gcloud app deploy --version=2
Services to deploy:

descriptor:                  [C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP\pom.xml]
source:                      [C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP]
target project:              [constant-gecko-425215-h8]
target service:              [servicio2]
target version:              [2]
target url:                  [https://servicio2-dot-constant-gecko-425215-h8.lm.r.appspot.com]
target service account:      [constant-gecko-425215-h8@appspot.gserviceaccount.com]


Do you want to continue (Y/n)?  y

Beginning deployment of service [servicio2]...
#=============================================================#
#= Uploading 0 files to Google Cloud Storage                 =#
#=============================================================#
File upload done.
Updating service [servicio2]...done.
Setting traffic split for service [servicio2]...done.
Deployed service [servicio2] to [https://servicio2-dot-constant-gecko-425215-h8.lm.r.appspot.com]

You can stream logs from the command line by running:
  $ gcloud app logs tail -s servicio2

To view your application in the web browser run:
  $ gcloud app browse -s servicio2

C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP>gcloud app deploy
Services to deploy:

descriptor:                  [C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP\pom.xml]
source:                      [C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP]
target project:              [constant-gecko-425215-h8]
target service:              [servicio2]
target version:              [20240602t190459]
target url:                  [https://servicio2-dot-constant-gecko-425215-h8.lm.r.appspot.com]
```

```
Símbolo del sistema

C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP>gcloud app deploy
Services to deploy:

descriptor:                  [C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP\pom.xml]
source:                      [C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP]
target project:              [constant-gecko-425215-h8]
target service:              [servicio2]
target version:              [20240602t190459]
target url:                  [https://servicio2-dot-constant-gecko-425215-h8.lm.r.appspot.com]
target service account:      [constant-gecko-425215-h8@appspot.gserviceaccount.com]


Do you want to continue (Y/n)?  y

Beginning deployment of service [servicio2]...
#=============================================================#
#= Uploading 0 files to Google Cloud Storage                 =#
#=============================================================#
File upload done.
Updating service [servicio2]...done.
Setting traffic split for service [servicio2]...done.
Deployed service [servicio2] to [https://servicio2-dot-constant-gecko-425215-h8.lm.r.appspot.com]

You can stream logs from the command line by running:
  $ gcloud app logs tail -s servicio2

To view your application in the web browser run:
  $ gcloud app browse -s servicio2

C:\Users\jamal\Downloads\holamundoGCP\holamundoGCP>
```

Ingresar la URL para acceder a mi aplicación https://servicio2-dot-constant-gecko-425215-h8.lm.r.appspot.com/

servicio2-dot-constant-gecko-425215-h8.lm.r.appspot.com

Food | Countdown Timer –... | Netcut Know who(n... | iLovePDF | Outils P... | Cronómetro Online... | Gmail | Google Traduction | Google Calendar - S... | Mi día - To Do | ChatGPT | Todos los marcadores

Hola Mundo!!